SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 1 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	27.10.2026	**Duration:**	60 minutes

Code of Conduct

I T. Tarun (1924260001) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T. Tarun _____

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

ONLINE FOOD WASTE MANAGEMENT SYSTEM

1. Study and Analyze the Scenario

1.1 Business Domain and Scenario

The Online Food Waste Management System belongs to the food sustainability and waste-reduction domain. Restaurants, hotels, cafeterias, grocery stores and households may have surplus edible food that could be safely redistributed instead of discarded. The proposed system provides an online platform for donors to list surplus food and for verified NGOs, charities and recipients to find and request available food, with collection status tracked through a centralized dashboard.

1.2 Existing Challenges

- Food donors may not have an efficient way to inform nearby charities about surplus food.

●Edible surplus food may be discarded because recipients are not identified in time.

●Manual coordination can delay food collection and increase the risk of spoilage.

●Paper-based records make it difficult to track donations, collections and food waste trends.

●Donors and recipient organizations may not receive timely updates about listings or collection changes.

1.3 Stakeholders

Stakeholder	Role / Responsibility
Food Donation Coordinator	Coordinates food donation activities and reviews reports.
System Administrator	Manages accounts, verifies organizations and configures the system.
Collection Coordinator	Coordinates collection issues and follows up on delayed pickups.
Recipient Organizations	View available food listings and donation notifications.
Donor / Restaurant Manager	Lists surplus food and coordinates pickup with recipients.
Notification Service	Delivers listing, request and collection updates.

1.4 System Objectives

●Allow donors to list surplus food with quantity, preparation date, expiry/use-by time and pickup location.

●Match available food listings with verified recipient organizations and their needs.

●Provide timely notifications about new listings, requests and collection deadlines.

●Track food requests, confirmations, pickups and completed donations.

●Generate reports to help measure food rescued and waste reduced.

2. Identify Functional and Non-Functional Requirements

2.1 Functional Requirements

●FR1 – Authentication: The system shall allow registered users to sign in with role-based access.

●FR2 – Food Listing: The system shall allow donors to publish food type, quantity, preparation time, expiry/use-by time, storage conditions and pickup location.

●FR3 – Listing Dashboard: The system shall display available food listings with quantities, locations and collection deadlines.

●FR4 – Food Search and Matching: The system shall allow recipients to search and filter listings by location, food type, quantity and availability.

●FR5 – Request Management: The system shall allow verified recipients to request available food and notify the donor.

●FR6 – Donation Confirmation: The system shall notify both parties when a donation request is accepted, rejected or updated.

●FR7 – Donation History: The system shall store timestamped listing, request and collection records.

●FR8 – Reporting: The system shall generate daily and monthly food donation and waste-reduction summaries.

●FR9 – User and Listing Management: Administrators shall verify accounts and manage food listings.

●FR10 – Notification and Status Management: Authorized users shall receive updates and track donation status through completion.

2.2 Non-Functional Requirements

●Performance: New food listings and request-status changes should appear promptly after submission.

●Security: Authentication, authorization and protected network communication shall be used.

●Reliability: The system shall preserve donation and collection records during temporary connection failures.

●Usability: The dashboard shall present food listings, requests and donation status in a clear and understandable format.

●Scalability: The design should support additional donors, recipient organizations and locations.

●Maintainability: Food listing, matching, notification and reporting components should be independently maintainable.

●Availability: The dashboard should remain accessible during normal operating periods.

●Data Integrity: Listings and donation records shall include timestamps, user identifiers and status history for traceability.

3. Identify Actors and Use Cases

3.1 Primary and Secondary Actors

Actor	Type	Major Interaction
Administrator	Primary	Manage users, verify organizations and manage listings
Donor / Restaurant Manager	Primary	Manage food listings, requests and reports
Collection Coordinator	Primary	Follow up on collection issues and update status
Recipient Organization	Primary	Search food listings, request donations and view notifications
Donor / Recipient Interface	Secondary	Submit listings and donation requests
Notification Service	Secondary	Deliver donation and collection notifications

3.2 Major Use Cases

No.	Use Case	Actor
1	Login and view dashboard	Administrator
2	Register users / verify organizations	Administrator
3	Create and publish food listing	Food Donor
4	Search available food listings	Recipient Organization
5	Match available food with recipient needs	System
6	Receive and respond to donation notifications	Donor, Recipient Organization
7	Generate food donation report	Donor / Restaurant Manager
8	Update collection status	Collection Coordinator

3.3 Use Case Diagram (Text-Based UML Layout)

Administrator → (Manage Users) (Verify Organizations) (Manage Listings)

Food Donor → (Create Food Listing) → (Publish Available Food)

Recipient Organization → (Search Food) (Request Donation) (Confirm Collection)

Donor / Recipient → (Track Donation Status) (Update Collection Status)

Donor / Recipient → (Receive Notifications) (View Donation History)

System boundary: All use cases above belong to the Online Food Waste Management System. Donors publish surplus food, verified recipient organizations request donations, and the system manages notifications, status tracking and reports.

3.4 Use Case Description

When a donor submits a surplus-food listing, the backend validates the required details and stores the listing. The system displays the listing to eligible recipient organizations. A recipient can request the food, and the donor can confirm the request and arrange collection. The system sends status notifications, records collection completion and allows administrators to review donation activity and generate food-rescue reports.

4. Prepare the Software Requirement Specification (SRS)

4.1 Introduction

The Online Food Waste Management System is a proposed web platform that connects surplus-food donors with verified recipient organizations to reduce avoidable food waste and support timely redistribution.

4.2 Problem Description

Food redistribution often depends on phone calls, informal messages or manual coordination. As a result, edible surplus food may be discarded, charities may not know what is available nearby, and collection may be delayed. An online system is required to publish food availability, coordinate requests, track collections and maintain reliable donation records.

4.3 Scope of the System

The system covers donor and recipient registration, food listings, search and matching, donation requests, confirmations, notifications, collection tracking, historical records and reporting. It does not independently guarantee food safety or replace applicable food-handling, storage, transport and legal requirements; users remain responsible for accurate information and safe handling.

4.4 Functional Requirements

The system shall implement FR1–FR10 described in Section 2.1, including authentication, food listing, search and matching, donation requests, status notifications, history and reporting.

4.5 Non-Functional Requirements

The system shall satisfy the security, performance, reliability, usability, scalability, maintainability, availability and data-integrity expectations described in Section 2.2.

4.6 Use Case Descriptions

Core workflows include creating food listings, searching available food, requesting donations, confirming collection, receiving notifications, reviewing donation history and generating reports. Each workflow requires appropriate permissions and accurate listing details.

4.7 Data Requirements

- User data: user ID, role, login credentials and contact preferences.

●Listing data: listing ID, donor ID, food type, quantity, preparation date/time, expiry/use-by time, storage requirements, pickup location and availability status.

●Donation data: donation ID, listing ID, recipient ID, requested quantity, request timestamp and collection status.

●Notification data: notification ID, recipient, message type, timestamp and read status.

●Collection data: collection ID, assigned recipient, pickup time, completion status and any issue reported.

●Report data: selected date range, quantities listed and collected, completed donations and estimated food waste avoided.

4.8 External Interface Requirements

●User Interface: Responsive browser dashboard for food listings, requests, notifications and reports.

●Listing Interface: Forms for entering food details, quantity, use-by time, storage conditions and pickup location.

●Communication Interface: Secure web/API communication between the application, database and notification service.

●Database Interface: Backend storage and retrieval of listings, users, donation requests and collection records.

●Notification Interface: Email, SMS or application alerts when a supported service is configured.

●Export Interface: Downloadable reports such as CSV or PDF.

4.9 System Features

●Online surplus-food listing and discovery

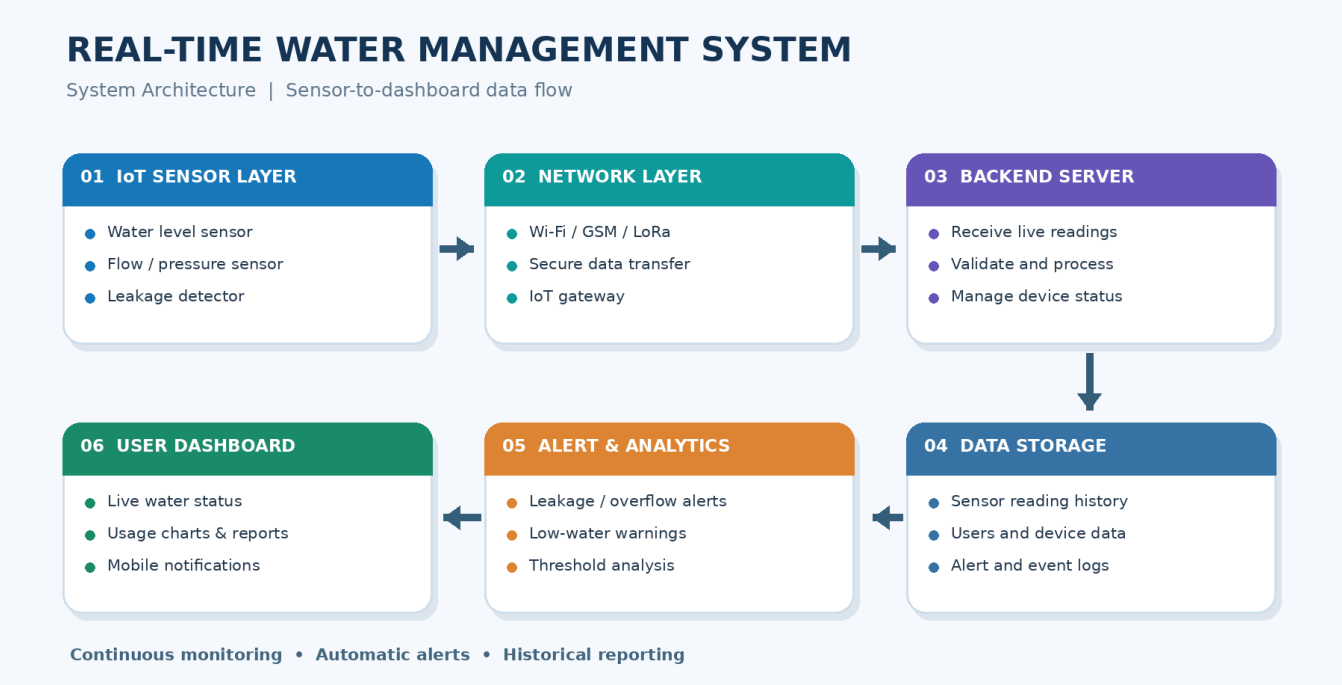
●Notifications for donation requests, confirmations and collection deadlines

- Food donation, collection and waste-reduction reports
- Listing availability and donation status tracking
- Role-based dashboards and user accounts
- Donation request, collection status and issue tracking

4.10 System Architecture Diagram

Layer	Flow / Responsibility
User Interface	Donor / Recipient Web Interface
Connectivity	Internet / Secure Communication
Processing	Application Server → Matching and Notification Services
Storage	Database → Food Listings, Requests and Collection Records
Presentation	Web Dashboard → Donation Reports and Notifications

Architecture flow: Donor / Recipient Web Interface → Application Server → Database + Matching and Notification Services → Dashboard / Reports.



5. Identify Assumptions and Constraints

5.1 Assumptions

- Donors provide accurate food descriptions, quantities, use-by times and storage information.
- Users have internet access and a suitable device to use the online platform.
- Authorized personnel can access the dashboard when required.
- Food listings are visible only while the food remains available and suitable for redistribution.
- Donors and recipients respond promptly to requests and collection notifications.

5.2 Constraints

- Technical: Internet outages or service failures may delay listing updates and notifications.
- Operational: Food must be handled, stored and transported safely, and collected within the stated time.
- Economic: Hosting, notification services, collection transport and maintenance must fit the project budget.
- Legal: Personal information and operational records must be handled according to applicable rules.
- Time: Implementation, integration and testing must fit the academic schedule.

6. Validate Requirement Completeness and Consistency

6.1 Completeness Validation

Verify that every stakeholder need is covered: food listings, search, donation requests, collection tracking, notifications, administration and reporting. Confirm that every listing and request has a responsible user and clear status.

6.2 Consistency and Clarity Validation

Use consistent quantity units, date/time formats and status labels across listings, records and dashboards. Make food expiry/use-by time and pickup conditions clear. Ensure that donors cannot modify administrator-only settings.

6.3 Feasibility and Test Validation

- Create sample food listings and verify that quantities, dates and locations display correctly.
- Test donation requests, accept/reject actions and notifications for each status change.
- Simulate an unavailable or expired listing and verify that recipients cannot request it.
- Test access restrictions for administrator, donor and recipient organization roles.
- Generate reports and compare listed, requested and collected quantities against stored sample records.

6.4 Suggested Improvements

Specify listing moderation rules, data retention periods, notification delivery targets and clear criteria for marking food unavailable. Add a documented process for expired listings, cancellations, missed pickups and reports of unsafe or inaccurate food information.

7. Conclusion

The proposed Online Food Waste Management System provides a structured approach to connecting surplus-food donors with recipient organizations, coordinating timely collection and reducing avoidable food waste. Its SRS defines the users, data, functions, interfaces, constraints and validation checks needed to guide reliable implementation.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional	Identifies most requirements with minor classification	Identifies only basic requirements; several	Requirements are incomplete, inaccurate, or poorly

(25%)	and non-functional requirements.	errors.	omissions.	classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25